

CivicNet ? App Encryption Declaration

****For:**** Apple App Store Connect ? App Encryption Documentation

****App name:**** CivicNet

****Bundle ID:**** 'com.sufyankamil.communityNet'

****App Store ID:**** 6761416586

****Developer / Account:**** Sufyan Kamil

****Repository:**** <https://github.com/sufyankamil/ai-civicnet>

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1. Purpose of encryption

CivicNet uses cryptography solely to protect ****private chat message content**** between users who have been matched for community help requests.

Encryption is used for:

? Confidentiality of chat message payloads

? Wrapping per-conversation symmetric keys for conversation participants

Encryption is ****not**** used for:

? Digital rights management (DRM)

? Proprietary or unpublished cryptographic protocols

? Concealing malware or circumventing security controls

2. Cryptographic algorithms

All algorithms are ****industry-standard**** and published by recognized standards bodies.

CivicNet does ****not**** implement proprietary or non-standard cryptography.

Use	Algorithm	Key size / params	Standard
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Chat message encryption	AES in CBC mode	256-bit key, 128-bit IV	NIST FIPS 197 / ISO/IEC 18033-3
Conversation key wrap	RSA-OAEP	2048-bit RSA, OAEP with SHA-256	PKCS #1 v2.2 / IETF RFC 8017
Transport to backend networking	TLS (HTTPS)	As negotiated by iOS	IETF TLS ? Apple OS

Legacy dual-read (migration only): older messages may still be decrypted with a former app-wide AES-256-CBC key during a limited migration window. New messages use per-conversation keys only.

3. Where cryptography is implemented

Layer | Implementation

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Message & key-wrap crypto | Client app (Flutter / Dart) via open-source libraries
'encrypt' and 'pointycastle'

Private key storage | iOS Keychain via 'flutter_secure_storage'

Network transport | Apple OS TLS / HTTPS to Supabase and Firebase

Cryptography for chat is implemented ****in the application****, in addition to encryption provided by the Apple operating system (HTTPS). It is ****not**** limited to Apple OS APIs such as CryptoKit for message payloads.

Source of truth in the codebase: 'lib/services/encryption_service.dart'.

4. What is encrypted

? ****Encrypted:**** chat message body ciphertext; per-conversation AES keys wrapped under each participant's RSA public key

? ****Not encrypted by custom app crypto:**** public profile fields intended for discovery; push notification bodies (generic text only)

Key lifecycle (summary):

1. Each device generates an RSA-2048 identity key pair; public wrap key is stored on the user profile.
2. When a conversation starts, a random 256-bit conversation key is generated.
3. That key is RSA-OAEP wrapped for each participant and stored server-side as opaque ciphertext.
4. Messages are AES-256-CBC encrypted with the conversation key; ciphertext is stored/transmitted as Base64(IV ? ciphertext).

5. Distribution

CivicNet is distributed via the ****Apple App Store****, including availability in ****France**** unless territories are later restricted by the Account Holder.

6. Classification statement

? Algorithms used are ****standard**** (AES, RSA-OAEP, SHA-256, TLS).

? No proprietary, unpublished, or non-standard cryptographic algorithms are used.

? Per Apple's matrix for industry-standard algorithms not limited to Apple OS encryption, this declaration supports App Store Connect export compliance review (French encryption declaration path when distributing in France).

? A US CCATS filing is ****not**** claimed as required for this standard-algorithm use case unless Apple's questionnaire explicitly requests one.

7. Contact

For questions from Apple App Review / export compliance regarding this documentation, contact the App Store Connect ****Account Holder**** for CivicNet ('com.sufyankamil.communityNet').

This document is the technical encryption declaration for App Store Connect upload. Export to PDF (Print ? Save as PDF) before uploading. If Apple also requires the official French ANSSI Cerfa form, see 'APP_STORE_ENCRYPTION.md'.